MASTER OF SCIENCE THESIS PROPOSAL

Spatiotemporal Modeling of Citizen Tree Watering for Equitable Climate Adaptation in Berlin

Climate Responses and Municipal Irrigation Priorities

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Abstract

This thesis examines how recorded citizen tree watering can complement conventional vulnerability assessments and inform municipal irrigation priorities in Berlin. Heat and drought can increase the need for tree care while limiting vegetation cooling. The Gieß den Kiez platform provides dated, geolocated watering records with tree identities and reported volumes. The current extract contains about 59,000 events across six seasons from 2020 to 2025, offering a basis for studying civic responses and their spatial distribution.

Four research questions guide three linked studies. Part II is the core study: it estimates how recorded events and volumes vary with heat, rainfall and antecedent drought, examines their temporal stability, and relates differences in logged care to environmental vulnerability, tree populations, socioeconomic conditions and pump access. It then tests what civic-care indicators add to conventional assessments. Part I supplies supporting evidence on vegetation cooling and its loss during dry conditions. Part III compares hypothetical municipal irrigation priorities under need-only and citizen-data-informed rules, including reporting-bias scenarios and explicit fairness safeguards.

The outcomes are logged events and litres, spatial differences between assessment layers, and changes in simulated allocation priorities. Morning land surface temperature (LST) supports the environmental assessment. These measures do not establish the causal effect of watering, total care received or actual unmet water demand. Candidate areas of unmet adaptation need will therefore be identified with uncertainty and treated as priorities for further assessment. The core design uses public data; pseudonymous user identifiers and municipal watering records are optional additions.

1 Introduction and Motivation

Heat exposure in Berlin is amplified by dense built form and night-time heat retention, and older residents and other vulnerable groups face disproportionate risk (Burkart et al., 2014; Gabriel and Endlicher, 2011; Masselot et al., 2023). Street and park trees are therefore both ecological infrastructure and a public-health adaptation resource.

Trees provide shade and evaporative cooling, but the evaporative component depends on water availability (Schwaab et al., 2021; Gobatti et al., 2025). Citizen watering may supplement municipal maintenance during dry periods. Its contribution depends on where residents act, what they record, and whether water is accessible. Tree age alone cannot establish whether a tree receives municipal care; that status requires district records.

Gieß den Kiez records voluntary watering, while Berlin's Baumkataster provides a register of more than 950,000 catalogued trees in the current extract. Linking these sources to weather, environmental conditions and social indicators makes it possible to examine recorded action alongside conventional measures of need. Public pump locations add a measure of access to a practical resource for tree care.

The thesis focuses on weather responses, spatial gaps in recorded care and their implications for public allocation. It does not estimate whether individual watering events measurably lower local LST. The scale assessment in Appendix A indicates that this signal is unlikely to be recoverable with the available satellite sampling. Low recorded care in a vulnerable area is treated as a possible gap requiring investigation, rather than proof that its trees receive no water.

The analysis combines geospatial data integration, hierarchical modelling with partial pooling, spatial and temporal validation, and transparent allocation scenarios. Fairness is an explicit policy choice evaluated alongside predictive performance and measurement uncertainty.

2 Background and Related Work

Conceptual framework

The thesis uses the IPCC risk framework (IPCC, 2014): risk depends on hazard, exposure and vulnerability, including sensitivity and adaptive capacity (Smit and Wandel, 2006). Environmental vulnerability is operationalised through thermal burden, vegetation and sealing conditions, with drought-related cooling loss as a supporting indicator. Social sensitivity and resource access are examined separately. Table 1 distinguishes these dimensions. Logged care measures observed action and reporting; it is not a direct measure of adaptive capacity or total irrigation.

Table 1. Framework elements and their operationalization.

Framework element	Operationalisation in this thesis	Source
Climate hazard	Daily maximum temperature, rainfall and antecedent dryness	DWD; drought alternatives
Environmental burden	Morning LST, sealing, vegetation and cooling loss	Landsat; Umweltatlas
Social sensitivity and resources	Social status, older-population share and pump access	MSS; population register; pump data
Tree-care need	Street-tree stock, age, genus and local conditions	Baumkataster; environmental layers
Recorded civic care	Events, litres, weather responses and persistence	Gieß den Kiez
Potential adaptation gaps	Joint pattern of conventional need and low recorded care, with uncertainty	Combined assessment; not observed unmet demand

Vegetation cooling and drought

Trees lower LST across European cities, although the relationship varies by regional climate (Schwaab et al., 2021; Alonzo et al., 2025). Soil moisture affects evaporative cooling (Gobatti et al., 2025), and thermal observations during drought show variation between species and built environments (Zandler and Samimi, 2024). Part I therefore separates tree canopy from low vegetation rather than relying on NDVI alone, while keeping surface cooling distinct from air temperature and human heat exposure.

Distribution of cooling loss during drought

Yan et al. (2025) examine unequal vegetation degradation and cooling during drought across US cities. Their results motivate a Berlin analysis of whether dry-condition cooling loss coincides with social disadvantage and existing thermal burden. This supporting study provides environmental

context for the civic-care analysis; the thesis does not depend on establishing a universal or uniquely European pattern.

Measuring adaptive capacity and civic climate response

Conventional vulnerability assessments often use social and environmental indicators observed at relatively infrequent intervals (Cutter et al., 2003). Behavioural studies provide a different perspective: Stechemesser and Wenz (2023), for example, examine unequal heat-related mobility responses in New York City. This thesis studies recorded collective care of shared green infrastructure. Its contribution will be assessed by the additional spatial information that this behavioural layer provides, rather than by treating disagreement with a conventional index as sufficient evidence of value.

Civic stewardship and participation inequality

Co-production depends on citizens' time, skills and access to resources (Ostrom, 1996). Digital reporting also varies across neighbourhoods, and platform design shapes participation (O'Brien, 2018; Kirklies et al., 2024). Differences in watering logs may therefore reflect actual care, platform awareness or reporting habits. Pump access and tree populations may influence the opportunity to act. The analysis estimates these associations and tests sensitivity to reporting assumptions; observational proxies alone cannot fully separate participation inequality from unrecorded care.

Decision support for tree irrigation

Berlin's QTrees project uses sensor and modelling approaches to estimate tree water needs (CityLAB Berlin), while Rebholz et al. (2026) describe irrigation decision support for urban green spaces. Part III examines the separate question of how incorporating recorded civic care changes municipal priorities. Need estimates and civic-care indicators remain distinct inputs, allowing their policy implications and uncertainty to be examined explicitly.

3 Research Gap

The proposal addresses four linked empirical gaps. A structured literature review in Month 1 will refine the scope of each contribution and check novelty claims.

- **Climate response.** Establish how logged events and volumes vary with heat, rainfall and antecedent drought, and whether these associations persist across seasons with changing platform activity.
- **Vulnerability and resources.** Examine whether environmental vulnerability coincides with differences in logged care after accounting for tree populations, socioeconomic conditions and pump accessibility.
- **Added spatial information.** Determine whether civic-care indicators reveal stable spatial differences beyond conventional vulnerability and tree-care indicators, while propagating uncertainty in observations and models.
- **Allocation consequences.** Quantify how need-only and citizen-data-informed priorities differ under a fixed hypothetical budget, and whether these differences persist under reporting-bias assumptions and fairness safeguards.

4 Research questions and analytical commitments

Structure and priorities

Study	Role	Research questions	Share of effort
Part II Civic care	Core	RQ1–RQ3: climate response, inequality and added information	≈ 50%
Part I Environmental context	Supporting	Supports RQ2–RQ3 with cooling-loss evidence	≈ 25%
Part III Allocation	Exploratory	RQ4: priorities, reporting bias and fairness	≈ 15%
Integration	Synthesis	Fundamental question and cross-scale uncertainty	≈ 10%

Fundamental research question

How can citizen tree-watering responses to heat and drought complement conventional vulnerability assessments to identify unmet adaptation needs and inform a more equitable allocation of municipal irrigation across Berlin's communities?

Research questions

Research question	What we would actually test
RQ1 — Civic climate response	How do recorded watering events and volumes change with heat, rainfall and antecedent drought, and how stable are these responses over time?
RQ2 — Vulnerability and resource inequality	Do areas with higher environmental vulnerability show different levels of logged care, and how are these patterns associated with tree populations, socioeconomic conditions and water-pump accessibility?
RQ3 — Added value over vulnerability assessments	What additional spatial information do recorded civic-care indicators provide beyond conventional vulnerability and tree-care indicators, and how robust is that information to measurement uncertainty?
RQ4 — Municipal allocation consequences	How do hypothetical need-only and citizen-data-informed irrigation priorities differ, and how sensitive are those differences to reporting bias and explicit fairness safeguards?

Hypotheses and evaluation criteria

The analysis plan will be registered before the main models are fitted, with existing exploratory analyses disclosed. RQ1 includes a directional hypothesis; RQ2–RQ4 are comparative questions whose direction and policy desirability remain open. Outcomes, contrasts, uncertainty summaries and sensitivity ranges will be specified in advance.

- **RQ1 hypothesis.** Logged events and volumes are expected to rise with heat and antecedent dryness and fall with rainfall. Report weather-response curves, standardised contrasts and 95% intervals, including possible non-linearity. Temporal stability is evaluated through year-specific estimates and earlier-versus-later season comparisons; non-significance alone will not establish stability.
- **RQ2 comparison.** Estimate differences in logged care across environmental vulnerability levels before and after accounting for tree stock, socioeconomic conditions and pump access. Report adjusted contrasts and uncertainty without presuming that more vulnerable areas receive less care.
- **Competing explanations.** Compare estimates after including baseline participation, calendar and year effects, civic-participation proxies and, if available, user concentration. Changes in

estimates indicate sensitivity to these factors; they do not identify reporting bias or demonstrate a causal mechanism.

- **RQ3 evaluation.** Assess spatial discordance, areas consistently classified as high vulnerability with low recorded care, and out-of-sample information about subsequent logged care. Require stability across measurement and modelling scenarios before treating a spatial pattern as informative. Weak correlation with an existing index alone is insufficient.
- **RQ4 evaluation.** Compare rank changes, top-priority overlap and simulated municipal shares by vulnerability and socioeconomic group. Repeat the comparison under reporting-bias scenarios and explicit safeguards. Allocation differences and their fairness implications are reported even when the two rules produce similar priorities.

Analytical mapping

RQ	Primary analysis and evidence
RQ1	Hierarchical event and volume models; weather-response curves; annual and early-versus-late contrasts; held-out seasons.
RQ2	Crude and adjusted care rates by environmental vulnerability; matched tree denominators; social conditions, tree composition and pump accessibility.
RQ3	Conventional versus civic layers; spatial discordance and stable joint classes; incremental information on subsequent logged care; uncertainty propagation.
RQ4	Need-only versus citizen-data-informed allocation under the same budget; ranking and group-share differences; reporting-bias scenarios and fairness safeguards.

5 Data Sources

The observation unit is the spatial or temporal support at which a value is used. Landsat LST is delivered on a 30 m grid from approximately 100 m native thermal sampling. Dataset versions, coverage periods and joins will be recorded. Later social, registry and pump snapshots are contextual proxies for earlier years unless historical versions can be obtained.

Dataset	Provider	Use	Unit and limit	Part
Landsat 8/9 C2 L2 LST	USGS via Earth Engine	Thermal burden and cooling	2017–2024 core; 30 m grid, ~100 m native; clear-sky mornings	I–III
Sentinel-2 L2A NDVI	ESA Copernicus	Dated greenness	2017–2024; 10 m aggregated to 30 m	I
Vegetation height 2020	Berlin Umweltatlas	Canopy and low vegetation	1 m raster; single-date structure	I–III
ECOSTRESS LST if available	NASA	Afternoon sensitivity	~70 m; irregular coverage	I
Imperviousness 2021 and 2016	Berlin Umweltatlas	Sealing and environmental burden	Polygon or raster; dated snapshots	I–III
Land use and waterbodies	Berlin Geoportal	Built form and water proximity	ISU5 and related spatial layers	I–III
Baumkataster	Berlin Geoportal	Tree stock, age, genus and joins	Tree records; coverage and history audited	II–III
Gieß den Kiez logs	CityLAB Berlin	Events, litres and continuity	Event-level; 2020–2025; launch and outages respected	II–III
Weather and heat warnings	DWD CDC	Temperature, rainfall and dry spells	Station-day; spatial representativeness checked	I–III

Dataset	Provider	Use	Unit and limit	Part
ERA5 and ESA CCI moisture	Copernicus and ESA	Drought alternatives	Coarse temporal indicators; no local moisture inference	I–II
MSS 2025	Berlin Senate	Social status	Planning area; later snapshot and exclusions	I–III
Environmental Justice Atlas 2023/24	Berlin Senate	Conventional burden benchmark	Planning area; overlapping components disclosed	II–III
Population register	Amt für Statistik BB	Population and older-age share	Planning area; annual where available	I–III
Public pump locations and status	Berlin Geoportal or districts	Water-resource accessibility	Point location; functionality and historical coverage audited	II–III
Election turnout 2021 and 2023	Berlin election office	Civic-participation proxy	Electoral districts crosswalked to planning areas	II
School and public holidays	Berlin Senate	Calendar controls	Day	II
Optional user identifiers	CityLAB request	Participation concentration	Pseudonymous event links if granted	II
Optional municipal watering records	District requests	Care status and scenario checks	Tree or street coverage if granted	III

6 Methodology

Study area and units of analysis

The study covers Berlin's 12 districts. The main behavioural unit is the planning-area-day using a harmonised set of 542 LOR Planungsräume, with 143 Bezirksregionen as a fallback. Part I aggregates satellite evidence to the same areas without treating adjacent 30 m thermal cells as independent. Part III uses tree-seasons and area-level allocations. Key spatial conclusions will be repeated at planning-area, Bezirksregion and district scale to assess aggregation sensitivity (Openshaw, 1984).

Data audit and signal construction

The current exploratory audit reports 111 exact duplicate logs, five events above 500 L, 56 coordinate swaps and 125 records remaining outside Berlin. Reported tree-ID and planning-area join rates are 98.95% and 99.90%, respectively. These figures will be reproduced against a versioned extract before modelling. Large volumes will be flagged and examined rather than automatically treated as errors. The audit will also:

- reconcile the current watered-tree counts of 11,087, 11,160 and 11,245 by documenting the filters and inclusion dates behind each figure;
- match each outcome to its denominator: the main rate analysis uses logs linked to catalogued street trees and the corresponding street-tree stock; an age 5–19 subgroup uses only events and trees in that age range;
- document tree-registry coverage, historical stock uncertainty, area boundary changes and actual platform observation dates; days before launch or during known outages are missing, not zero;
- audit pump coordinates, functionality and historical coverage; record the outcome of optional CityLAB and district requests, with predefined proxy analyses if access is unavailable.

The exploratory summer series is strongly overdispersed and varies substantially by year: mean daily events are reported as 121.6 in 2020, 44.7 in 2021, 93.1 in 2022, 39.3 in 2023, 19.4 in 2024

and 24.7 in 2025. These summaries motivate a count model with overdispersion, year effects and seasonal controls. They also make temporal stability a substantive part of RQ1. Provisional counts will be checked after the final sample definition is fixed.

Part I Environmental vulnerability and cooling loss

Outcome and role. Landsat 8/9 morning LST on valid clear-sky summer overpasses in 2017–2024 provides thermal context; 2013–2024 may be used for descriptive maps. Cooling-loss estimates support RQ2 and RQ3. Inference is limited to the observed clear-sky regime, and the weather distribution of sampled versus all summer days will be reported.

Vegetation. Canopy fraction and low-vegetation fraction per cell come from the 2020 vegetation-height layer; Sentinel-2 NDVI from the nearest valid scene (window length stored) captures dated greenness. Because heat and water stress lower NDVI, lagged NDVI is used as a reverse-causality check.

Drought. A city-level temporal anomaly from ESA CCI soil moisture, with ERA5 and DWD-based SPEI as alternatives. All spatial variation comes from vegetation and built form, never from the coarse moisture grid. Dry and wet are defined as anomaly terciles, with quartile and continuous versions as sensitivity.

Model. A hierarchical model estimates the vegetation–LST association and its variation with drought and social context. The model includes the relevant lower-order interactions so that the three-way term is interpretable:

$$\mathrm{LST}(c,t) = \alpha(t) + \beta_1 V(c) + \beta_2 V(c)D(t) + \beta_3 V(c)S(j) + \beta_4 D(t)S(j) + \beta_5 V(c)D(t)S(j) + \beta_6 S(j) + \gamma' X(c) + u(j) + v(j)V(c) + w(j)V(c)D(t) + \varepsilon(c,t)$$

Here c denotes a cell, t an overpass date and j a planning area. Date effects α(t) absorb common weather and sensor conditions; V represents canopy and low vegetation as separate terms; D is the temporal drought anomaly; S is social status; and X includes sealing, built density and distance to water. Cooling loss is derived from dry-versus-wet vegetation contrasts, and area-level loss also accounts for vegetation fraction. Uncertainty is propagated to subsequent comparisons. Residual spatial dependence is assessed with Moran statistics (Anselin, 1995) and addressed with spatial effects where needed.

Checks. Spatially blocked validation assesses leakage (Roberts et al., 2017). Sensitivities use continuous drought anomalies, alternative drought series, lagged NDVI and available afternoon ECOSTRESS observations. Sealed cells provide a comparison for general drought-related thermal patterns; a vegetation interaction cannot be identified in a sample with no vegetation variation. Correlated vegetation and built-form predictors will be checked before interpreting individual coefficients.

Part II Civic climate response and resource inequality

Outcomes and observation window. RQ1 models daily logged event counts and total reported litres from April to September in 2020–2025, with 2020 beginning at the actual platform start date. City-wide summaries include valid records; rate models use the matched street-tree sample defined in the audit. Weather associations are estimated from within-season variation with calendar controls. A missing log does not reveal whether care occurred without being reported.

Participation and sparse areas. An area-level model first describes whether any watering is recorded, using tree stock, vulnerability, social conditions and pump access. The daily model includes observed zero days in areas with eligible trees. In areas with no recorded events, local weather-response slopes are largely determined by partial pooling; they will be flagged as weakly identified rather than interpreted as measured zero response. Results among participating areas provide a sensitivity analysis.

Daily event model. A hierarchical negative-binomial model includes an offset for matched street-tree stock, area intercepts and area-specific responses to daily maximum temperature and antecedent dry-spell length. Rainfall is entered separately, with prespecified lag windows and non-linear terms where supported. Controls include weekday, school and public holidays, year and day of season; heat-warning terms are examined for collinearity with temperature. Weather-by-year interactions and partially pooled annual slopes test stability. Environmental burden, social status, older-population share, population density and pump access explain variation in baseline care and, where estimable, weather responses.

Reported volumes. A two-part model estimates the probability of positive daily volume and the amount conditional on a positive total, using a lognormal or Gamma specification selected within training data. Combining both parts yields expected logged litres per day and weather contrasts. Event counts, litres and distinct trees watered are reported together. Sensitivities flag or exclude implausible volumes, vary drought and rainfall definitions, and assess missing or heaped volume entries.

Resource inequality. RQ2 compares crude and adjusted care rates across environmental vulnerability levels. Tree counts, age and genus composition are considered alongside socioeconomic conditions. Pump accessibility is measured by tree-to-pump distance and the share of trees within prespecified distance bands; network distances are preferred where feasible, with straight-line distances as a sensitivity. Available functionality records distinguish usable from mapped pumps. Cross-sectional pump access and civic-participation proxies are associations, not causal explanations or validated measures of digital access.

Estimation and validation. Initial fits use glmmTMB or INLA at Bezirksregion scale, followed by brms/Stan where feasible (Bürkner, 2017), with partial pooling for sparse areas (Gelman and Hill, 2007). Checks include temporal dependence, posterior predictive fit, held-out years and spatially blocked folds. Year-specific contrasts and a 2020–2022 versus 2023–2025 comparison address RQ1 stability. If user identifiers are available, distinct users and exclusion of highly active users assess concentration; otherwise distinct watered trees and concentration across trees serve as limited proxies.

Interpretation. Civic climate response describes weather-associated variation in recorded care. It does not measure residents' own protection from heat or identify the causal effect of weather, socioeconomic conditions or pump provision. Comparisons describe places, not individual volunteers. Contemporary social and infrastructure snapshots cannot establish historical conditions throughout 2020–2025.

Added spatial information beyond conventional assessments

Baseline and civic layers. RQ3 benchmarks a conventional assessment combining environmental burden, social sensitivity and tree-care indicators such as age, genus, sealing and stock. Pump accessibility is included as a resource indicator. Civic layers describe logged events and litres per tree, weather-response contrasts and continuity of recorded care across seasons. Indicators retain separate meanings; any composite weighting is declared and varied.

Spatial contribution. Compare rankings and joint classifications, including high conventional need with low recorded care. Report Spearman correlations, changes in priority groups and the probability that each area remains in a category across uncertainty draws. Discordance is interpreted alongside tree coverage, sparse observations and reporting assumptions. The association between cooling loss and civic response is a supporting comparison within RQ3, not a separate research question.

Incremental information. Compare conventional-only models of subsequent logged care with models that also include civic indicators from earlier seasons. Use temporal and spatial hold-outs

and assess calibration and predictive error. This checks whether civic records contain persistent information about recorded behaviour; it does not validate unmet need or improved ecological outcomes. Independent municipal, sensor or tree-condition observations, if obtained, provide a separate external check.

Measurement uncertainty. Propagate uncertainty in care responses and cooling estimates, perturb reported volumes and tree denominators, and vary assumptions about missing pumps, unrecorded care and reporting differences between areas. Present stable patterns, unstable patterns and areas with insufficient evidence separately. Added information is claimed only within the measurement and validation limits established by these checks.

Part III Municipal allocation consequences

Population and need. RQ4 uses catalogued street trees aged 5–19 years as a clearly defined scenario population, with wider age ranges as sensitivities. This restriction does not establish absence of municipal care. A conventional need score combines age, genus drought sensitivity and sealing, with explicit environmental and social weights. Seasonal water requirements are scenario inputs; sensor-based estimates or district records can refine them if available.

Care prediction. Models estimate the probability of recorded watering and logged litres conditional on watering. Predictors include tree attributes, local environment, pump access and area context. Previous-season care features are constructed only from information available at the allocation date. Regularised logistic regression and gradient boosting are compared with simple historical baselines; a positive-volume model supplies the conditional litres estimate.

Validation. Spatially blocked and temporal hold-outs assess predictions, with a final temporal test on 2024–2025 after training on 2020–2023. Hyperparameters and probability calibration are fitted within training folds. Tree or neighbouring-tree care from a held-out spatial block is excluded when testing transfer to new areas. AUROC, PR-AUC, Brier scores, calibration plots and volume error are reported (Niculescu-Mizil and Caruana, 2005). No future logs enter an earlier allocation scenario.

Allocation rules. Need-only priorities use conventional need inputs. A citizen-data-informed scenario subtracts an assumed effective civic contribution from seasonal requirements before ranking residual need. Let R be assumed water requirement, N a dimensionless need weight, $\hat{C}$ predicted logged litres (probability of a log × conditional litres), and a a scenario factor translating logged care into effective contribution:

$$\text{Need-only score} = N \times R$$
$$\text{Citizen-data-informed score} = N \times \max(0, R - a \times \hat{C})$$

The factor a is varied to represent uncertain reporting completeness, volume accuracy and effective delivery; it is not estimated as a correction without independent evidence. Care is capped at the assumed requirement. Both rules allocate the same hypothetical municipal water budget, with area totals derived from tree-level allocations. Report rank correlations, top-50 overlap, rank changes and municipal shares by vulnerability and socioeconomic group. These are simulated priorities and residual requirements, not measured unmet water demand or operational delivery plans.

Fairness and bias scenarios. Compare the unconstrained citizen-data-informed rule with safeguards: a guaranteed share of the need-only allocation for high-vulnerability areas, a cap on reductions attributed to civic care, and no care-based reduction where evidence is too uncertain. Vary safeguard thresholds and area-specific reporting factors, including lower assumed reporting in disadvantaged areas. Report who gains or loses municipal support, whether burden shifts to volunteers, and which priority changes remain stable. Missing municipal records and uncertain pump access enter separate sensitivity scenarios.

Integration and policy interpretation

Part I supplies environmental context; Part II supplies observed-care levels, weather responses and persistence; Part III evaluates allocation consequences. Their intersection identifies areas for further assessment, especially where conventional need is high and recorded care is consistently low. Civic indicators remain a separate evidence layer, and low recorded care is never equated with low willingness or capacity. Need-only and citizen-data-informed results are presented together so that the assumptions about civic effort and fairness remain visible.

Robustness and uncertainty

Primary outcomes and contrasts are specified before main-model estimation. Sensitivity analyses vary volume filters, tree denominators, drought measures, pump functionality, spatial scale and reporting assumptions. Posterior draws or appropriate resampling propagate uncertainty into maps and allocation comparisons. Any families of secondary frequentist tests use multiplicity control; Bayesian intervals are reported as such. Uncertain areas and unstable rankings remain visible rather than being forced into definitive categories.

7 Expected Contributions

- **Civic climate response.** Estimates of how logged events and volumes vary with heat, rainfall and antecedent drought, including temporal stability and the limits imposed by changing platform use.
- **Vulnerability and resource inequality.** Spatial comparisons linking recorded care to environmental conditions, tree populations, socioeconomic circumstances and pump accessibility, supported by a separate cooling-loss analysis.
- **Added information and allocation consequences.** An uncertainty-aware assessment of what civic-care indicators add to conventional assessments, followed by transparent simulations of municipal priorities under reporting bias and fairness safeguards.
- **Reproducible pipeline.** Documented integration of satellite, weather, tree-registry, pump and citizen-log data, with reusable models, validation procedures and an indicator dictionary.

The behavioural framework can be adapted to other places with comparable records. Future improvements in thermal observations or municipal data can refine the environmental and allocation components without changing the distinction between recorded care, inferred need and policy choice.

8 Work Plan and Timeline

Phase	Period	Tasks	Deliverable or decision
1 Setup and audit	Month 1	Literature review; source audit; pump and tree coverage; data requests; register analysis plan	Protocol and documented sample definitions
2 Analysis panels	Months 1–2	Area-day events and litres; weather; matched tree stock; pump access; satellite panel	Versioned panels and conventional indicators
3 Core civic analysis	Months 2–4	RQ1 climate response and stability; RQ2 vulnerability and resource associations	Care-response estimates with uncertainty
4 Environmental context and added value	Months 3–4	Cooling model; RQ3 joint spatial patterns and held-out information tests	Combined assessment; fix RQ4 scenario scope
5 Allocation scenarios	Month 5	RQ4 baseline and informed priorities; reporting bias; fairness safeguards	Allocation comparisons and sensitivity results

Phase	Period	Tasks	Deliverable or decision
6 Synthesis	Month 6	Cross-scale checks; answer fundamental question; writing and code release	Submitted thesis and documented outputs

9 Risks and Mitigation

Risk	Likelihood	Mitigation
Slow or unstable hierarchical models	Medium	Fit simpler or aggregated models first; reduce optional interactions while retaining core comparisons.
Missing user IDs or municipal records	High	Use declared concentration proxies and scenario bounds; do not infer municipal status from age alone.
Incomplete pump status or history	Medium	Compare mapped versus confirmed usable pumps; vary distance metrics; flag temporal mismatch.
Weak climate signal or changing logging	Medium	Report weak or unstable responses; retain care-level and uncertainty comparisons.
Uncertain cooling estimates	Medium	Keep conventional environmental layers as the baseline; treat cooling loss as supporting evidence.
Time overrun	Medium	Simplify Part I and care-prediction models; retain all four questions, bias checks and fairness comparisons.

10 Expected Outputs and Publication Strategy

- **Thesis.** Approximately 70–80 pages covering the conceptual framework, RQ1–RQ3 as the core results, supporting environmental analysis, RQ4 scenarios, limitations and synthesis. Appendices document data audits, scale sensitivity and model diagnostics.
- **Atlas and data layers.** Planning-area maps of vulnerability, logged-care levels, climate responses and persistence; joint classifications with uncertainty; and allocation differences under alternative assumptions. No map is labelled as verified unmet need.
- **Code.** Version-controlled pipeline and documentation.
- **Primary paper.** Civic climate response, resource inequality and added information beyond conventional assessments, with potential submission to Global Environmental Change, Landscape and Urban Planning or npj Urban Sustainability.
- **Optional second paper.** Drought-related vegetation cooling and its spatial distribution, conditional on the supporting analysis yielding a sufficiently clear contribution.

11 Ethical Considerations and Limitations

Data ethics

Public logs record tree IDs, dates, locations and volumes, not volunteer identities. If pseudonymous user IDs are provided, they are used only in aggregate, never linked to other data, and stored under a data-use agreement. Outputs avoid maps or tables that could re-identify individuals, for example areas with a single active volunteer. Socio-economic data are used only in published aggregate form. Ethics requirements will be confirmed with the university before analysis.

Allocation raises a distributive question: reducing public irrigation where residents volunteer can shift maintenance burdens onto those communities. The proposal therefore evaluates safeguards alongside both allocation rules. It also tests whether under-reporting changes apparent priorities. No rule is presented as uniquely fair; assumptions and trade-offs are stated explicitly.

Methodological limitations

- **Surface, not felt heat.** LST at ~10:30 is not near-ground air temperature or night-time heat, which matter most for health.
- **Incomplete observation of care.** Unlogged and municipal watering remain partly or wholly unobserved. Reported litres do not directly measure water reaching roots or meeting physiological demand.
- **Descriptive response.** Civic climate-response indicators are associational and are not direct measures of adaptive capacity. Area-level relationships cannot be attributed to individuals.
- **Clear-sky selection and coarse drought.** Part I covers fair-weather days and uses a city-level drought series.
- **Temporal mismatch.** Single-date vegetation, pump and tree-registry snapshots and later social indicators may not represent every study year; changes and coverage uncertainty enter sensitivity analyses.
- **Registry and infrastructure coverage.** Green-space trees are incompletely catalogued, vitality information is limited, and a mapped pump does not guarantee that it is usable.
- **Scale and policy assumptions.** Aggregation may change results. Simulated allocation depends on water requirements, reporting assumptions, budget and fairness thresholds, and does not establish ecological or social outcomes.

Appendix A Satellite detectability of individual watering effects

Dose. Using 500 L per growing season as an illustrative benchmark for a young tree, the current exploratory sample has a median of 65 logged litres per tree-season; 42.0% of logged tree-seasons reach 100 L, 15.9% reach 250 L and 6.3% reach 500 L. These are recorded-volume summaries, not estimates of the proportion of physiological need met.

Spatial concentration. On a 30 m grid, 7,174 cells contain watered trees, with a median of one per cell (90th percentile 3, maximum 11). On a 100 m grid, 4,205 cells, median two (90th percentile 6, maximum 29).

Energy-budget illustration. With latent heat of 2.45 MJ kg^{-1} and assumed surface coupling of about 20 W m^{-2} K^{-1}, the implied surface-temperature change if 30 L evaporates within the stated window is:

Case	If evaporated over 6 h	If evaporated over 24 h
30 L over one 100 m cell	0.34 W m^{-2} → 0.017 K	0.085 W m^{-2} → 0.004 K
30 L over one 30 m cell	3.8 W m^{-2} → 0.19 K	0.95 W m^{-2} → 0.047 K
Densest 100 m cell (29 trees, one day)	→ 0.49 K	→ 0.12 K

The six-hour and 24-hour cases are illustrative upper bounds for the stated evaporated dose. Poured water can infiltrate and be transpired over subsequent days, so logged volume does not establish an instantaneous latent-heat flux at the morning satellite overpass.

Detectability. Landsat C2 L2 LST has 1–2 K RMSE against ground truth, but much of that is bias that cancels in within-pixel comparisons, so the relevant noise is the pixel-level precision of a change. Even assuming an optimistic 0.3 K, a 0.05 K effect would require roughly 280 independent paired treated–control observations at matched times. Individual cells offer only 10–20 clear-sky summer overpasses per year, watering is rarely timed just before an overpass, and watered trees are not randomly chosen.

At the observed densities and available sampling, this assessment does not support a satellite-based causal test of individual watering effects. A designed experiment with controlled timing and concentrated watering would be needed to revisit that question. The calculation serves to bound the thesis scope, rather than to imply that watering has no cooling or ecological benefit.

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Approval and Signatures

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